

Introduction to Data Science Final Project

Jillian Melbourne

October 2025

1 Introduction

The RNA-seq count and meta data utilized in this project is from breast cancer patients in The Cancer Genome Atlas Breast Invasive Carcinoma (TCGA-BRCA) Cohort [1]. Data collection for this project was funded jointly by the National Cancer Institute and the National Human Genome Research Institute in an effort to connect cancer genotypes and phenotypes.

The first gene I explored in my analyses was CFLAR. I selected CFLAR because it is implicated in apoptosis and immune response regulation. In cancer, elevated expression of CFLAR is correlated with resistance to therapy [2].

The second gene I selected was TSPAN6, which is involved in extracellular signalling to recruit immune cells to the tumor microenvironment [3].

2 Methods

2.1 ggplot2

To generate, my histogram, scatterplot, violin plots, and cumulative distribution function figure, I used ggplot2 functions. I also used the ggplot2 `themeset()` function to automatically set the themes for all subsequent plots. This allowed me to standardize the font, font size, label position, background, and gridmarkers across the plots I generated with ggplot2.

2.2 ggpubr

I used ggpubr to add the regression line, correlation coefficient, and p value in my scatterplot.

2.3 biomRT

Audrey recommended I replace the Ensembl Gene IDs with the gene symbol to improve the readability of my heatmap. Inspired by Chayce's work, I used biomRT to rename my rownames as the corresponding gene symbol prior to generating my heatmap.

2.4 ComplexHeatmap

I used ComplexHeatmap to generate my heatmap and annotations.

3 Results

3.1 Summary Statistics

Table 1. Summary Statistics for CLFAR and TSPAN6

Gene	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	SD
CFLAR	664	3346	4799	5442	6720	21182	3040.0
TSPAN6	69	1562	2700	3208	4144	40780	2510.3

3.2 Histogram

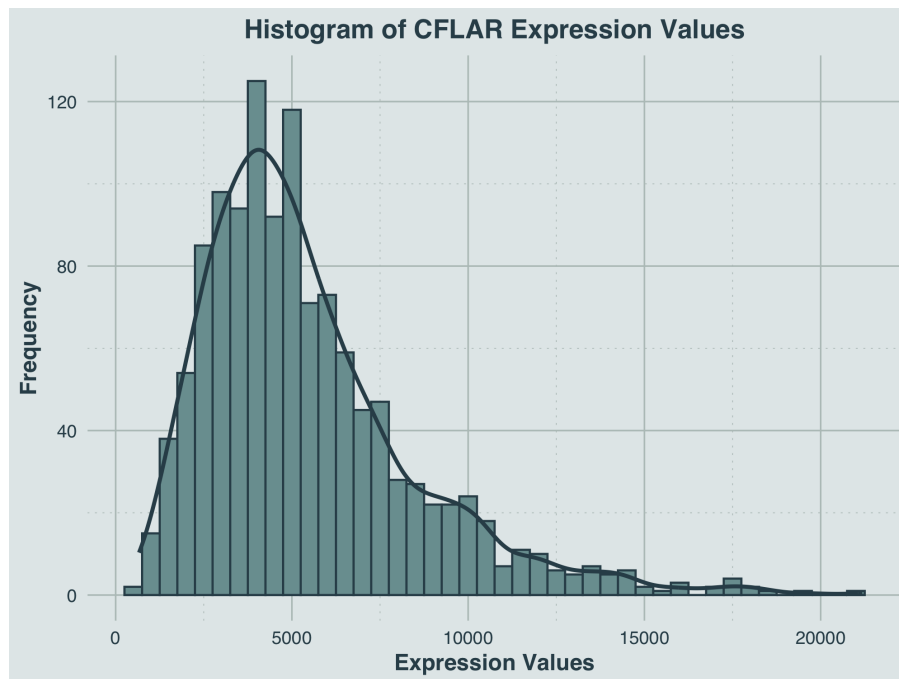


Figure 1: Histogram of CFLAR Expression

This histogram displays the frequency distribution of CFLAR expression values, with a fitted density curve shown in dark blue. The distribution is

right-skewed, with a mean expression level of approximately 5,000.

3.3 Scatterplot

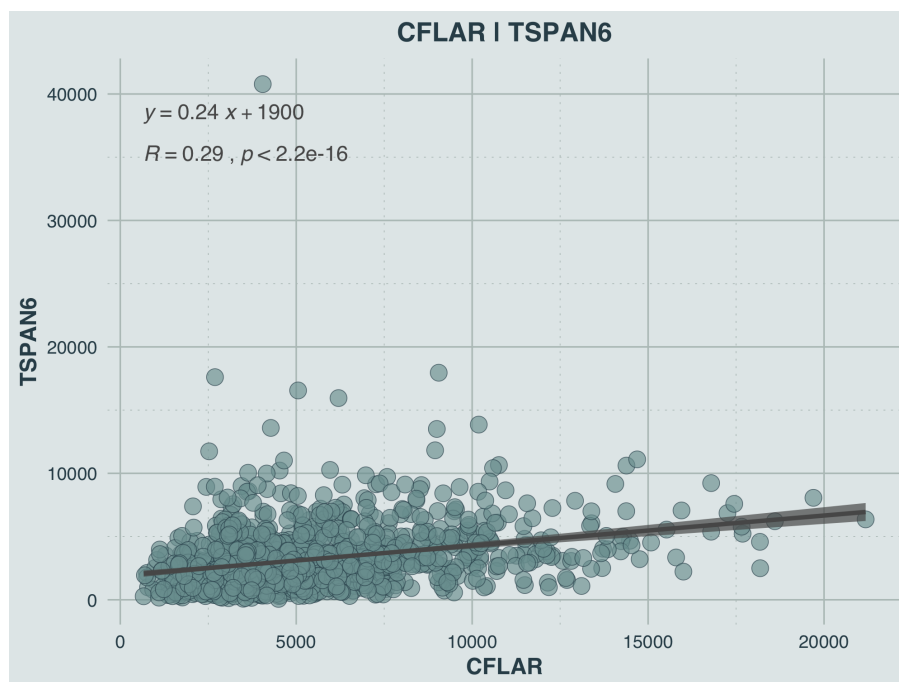


Figure 2: Scatterplot of CFLAR and TSPAN6 Expression

This scatterplot illustrates the genes expression values of CFLAR on the x-axis and TSPAN6 on the y-axis, with a regression line and confidence intervals in dark grey. In the top left corner of the plot, there is the formula for the regression line, the correlation coefficient, and the p-value. Within this cohort, there appears to be a significant, although very small, positive relationship between the expression of CFLAR and TSPAN6.

3.4 Violin Plot

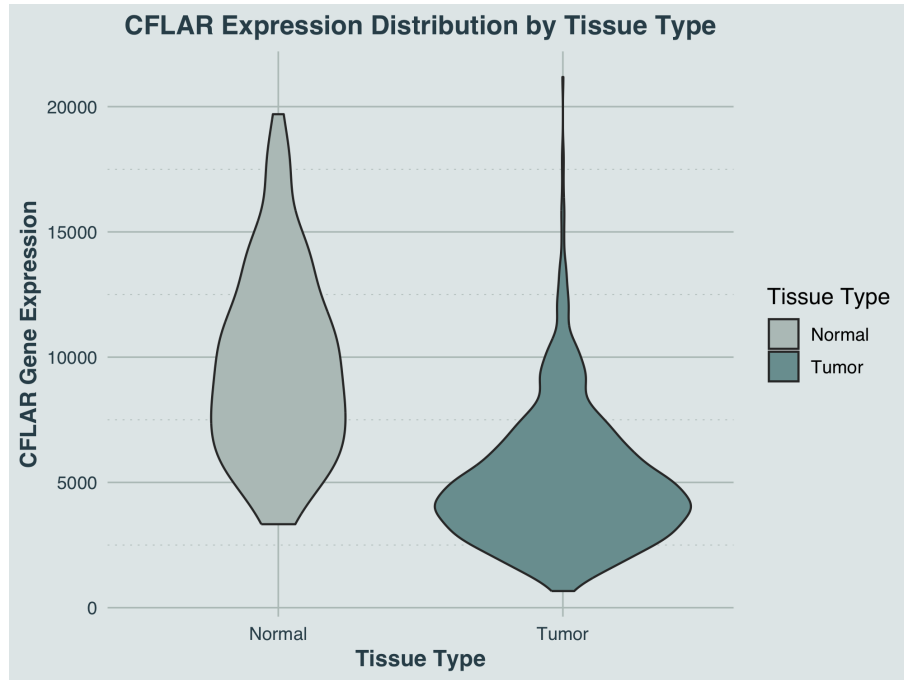


Figure 3: Violin Plot of CFLAR by Tissue Type

This violin plot has tissue type on the x-axis and CFLAR expression on the y-axis, with normal tissue in light grey and tumor tissue in dark teal. The distribution of the CFLAR gene expression for tumor tissue has a slightly lower mean compared to normal tissue. This is contrary to what I would expect because CFLAR contributes to tumor formation by inhibiting apoptosis. The observed trend may reflect characteristics of our cohort, such as the specific cancer types included or the relative proportions of normal and tumor samples in the dataset.

3.5 Heat Map

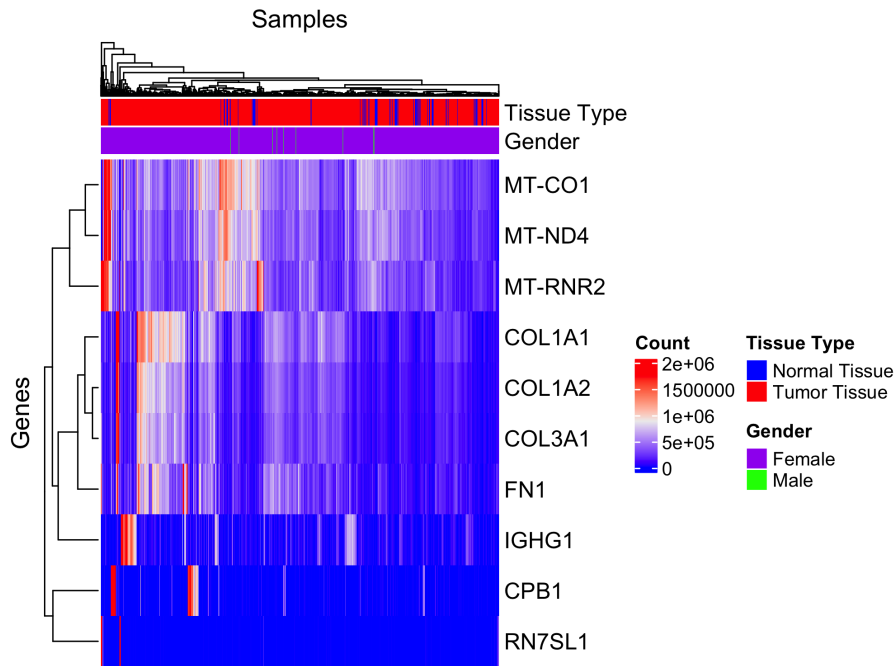


Figure 4: Heatmap of Gene Expression Across Tissue Types and Genders

This heatmap displays gene expression levels for the genes listed on the y-axis, where red indicates high expression and blue indicates little to no expression. The top annotations denote tissue type and gender. The genes shown were selected for having the highest variability across the cohort. Within our sample, genes such as MT-CO1 exhibit higher expression levels than RN7SL1. The annotations also indicate that most samples are derived from tumor tissue and that the majority of participants are female.

3.6 Cumulative Distribution Function

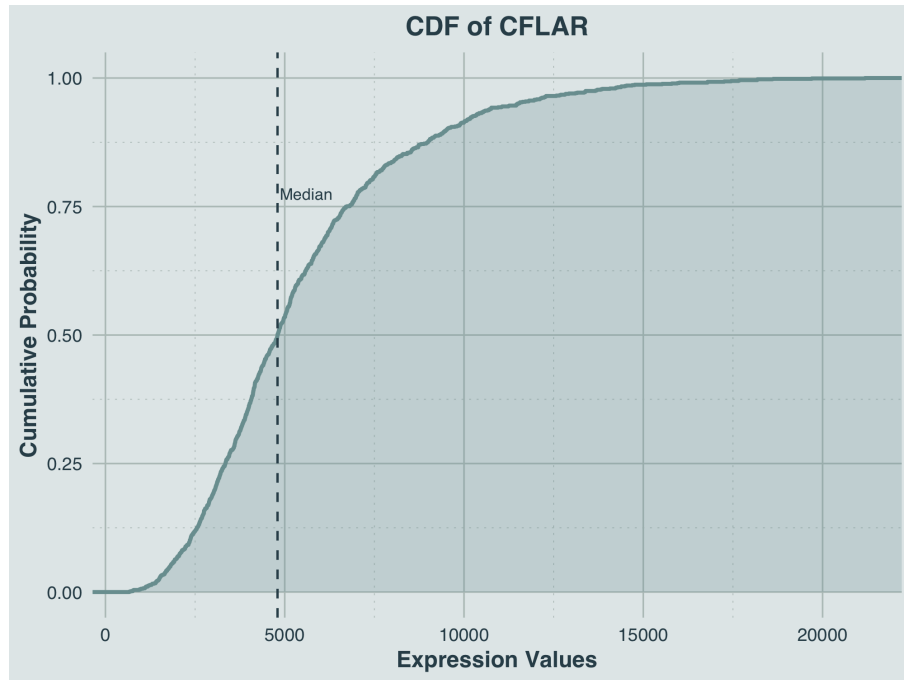


Figure 5: Cumulative Distribution Function of CFLAR

This Cumulative Distribution Function demonstrates the probability of having CFLAR expression below a certain value. For example, there is a 50 percent chance of having an expression value below 5,000. The shaded region is the area under the curve and the dashed line on this figure is the median.

4 References

1. Lingle, W., Erickson, B. J., Zuley, M. L., Jarosz, R., Bonaccio, E., Filippini, J., Net, J. M., Levi, L., Morris, E. A., Figler, G. G., Elnajjar, P., Kirk, S., Lee, Y., Giger, M., Gruszauskas, N. (2016). The Cancer Genome Atlas Breast Invasive Carcinoma Collection (TCGA-BRCA) (Version 3) [Data set]. The Cancer Imaging Archive.
2. Zhang H, Li X, Lin L. Biological Functions and Clinical Implications of CFLAR: From Cell Death Mechanisms to Therapeutic Targeting in Immune Regulation. *J Inflamm Res.* 2025 Apr 9;18:4911-4928. doi: 10.2147/JIR.S519885. PMID: 40224389; PMCID: PMC11994107.
3. Molostvov G, Gachechiladze M, Shaaban AM, Hayward S, Dean I, Dias IHK, Badr N, Danial I, Mohammed F, Novitskaya V, Paniushkina

L, Speirs V, Hanby A, Nazarenko I, Withers DR, van Laere S, Long HM, Berditchevski F. Tspan6 stimulates the chemoattractive potential of breast cancer cells for B cells in an EV- and LXR-dependent manner. *Cell Rep.* 2023 Mar 28;42(3):112207. doi: 10.1016/j.celrep.2023.112207. Epub 2023 Mar 3. PMID: 36867531.